# Problem 1: Breast Cancer Classification

**Dataset:** https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

### Objective

Predict whether a tumor is:

- Malignant (M)
- Benign (B)

### Tasks

1. Load the dataset.
2. Drop unnecessary columns (e.g., id, Unnamed: 32).
3. Convert target labels (M → 1, B → 0).
4. Split the dataset (80/20).
5. Standardize the features.
6. Train an SVM with a **Linear Kernel**.
7. Evaluate using:
    - Accuracy
    - Precision
    - Recall

# Problem 2: Mushroom Classification

**Dataset:** https://www.kaggle.com/datasets/uciml/mushroom-classification

### Objective

Predict whether a mushroom is:

- Edible
- Poisonous

### Tasks

- Handle categorical variables.
- Encode all features.
- Train Linear SVM.
- Evaluate performance.
- Compare with Decision Tree.

# Problem 3: Bank Marketing

**Dataset:** https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset

### Objective

Predict whether a customer will subscribe to a term deposit.

### Tasks

- Handle categorical variables.
- Feature scaling.
- Train Linear SVM.
- Print evaluation metrics.

## Problem 4: Heart Disease Prediction

**Dataset:** https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

### Objective

Predict whether a patient has heart disease.

### Tasks

Train four SVM models:

- Linear
- Polynomial
- RBF
- Sigmoid

Compare

- Accuracy
- Precision
- Recall

## Problem 5: Customer Churn Prediction

**Dataset:** https://www.kaggle.com/datasets/blastchar/telco-customer-churn

### Objective

Predict whether a customer will churn.

### Tasks

- Handle missing values.
- Encode categorical columns.
- Scale numerical columns.
- Train SVM.
- Tune hyperparameters.

## Problem 6: Credit Card Fraud Detection

**Dataset:** https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

### Objective

Detect fraudulent transactions.

## Tasks

- Handle imbalanced data. ( there is a parameter in SVC to address this )
- Train SVM.
- Use:
    - Precision
    - Recall
    - F1-score